Attention Mechanisms for Graph Learning

Ada Lovelace, Alan Turing

Abstract

We study attention mechanisms for graph neural networks and evaluate them on citation benchmarks with reproducible settings.

1 Introduction

Graph neural networks aggregate neighbourhood information. This paper introduces an attention based aggregation scheme.

Paragraph 1: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 2: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 3: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 4: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 5: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 6: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 7: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 8: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 1: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 2: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 3: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 4: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 5: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 6: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 7: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

Paragraph 8: This two column section discusses graph neural network message passing, aggregation and evaluation with enough detail for layout detection. Comma separated clauses, numbers 1, 2, 3, and citations make it realistic.

4 Experiments

Table 2: Accuracy comparison

Model	Accuracy	F1
GNN	0.91	0.88
CNN	0.85	0.80
MLP	0.72	0.69

The GNN model achieves the best accuracy on the benchmark.

5 Architecture

Figure 3: Architecture overview

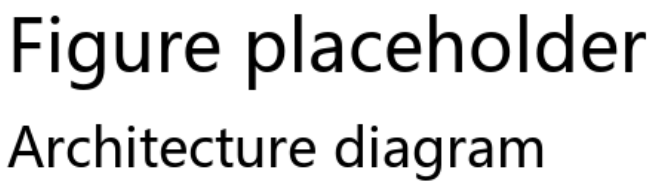


Figure 3 shows the overall architecture of the proposed model.

6 Discussion

Page 6 continues the discussion of limitations, reproducibility and future work for graph learning systems.

7 Discussion

Page 7 continues the discussion of limitations, reproducibility and future work for graph learning systems.

8 Discussion

Page 8 continues the discussion of limitations, reproducibility and future work for graph learning systems.

9 Discussion

Page 9 continues the discussion of limitations, reproducibility and future work for graph learning systems.

10 Discussion

Page 10 continues the discussion of limitations, reproducibility and future work for graph learning systems.